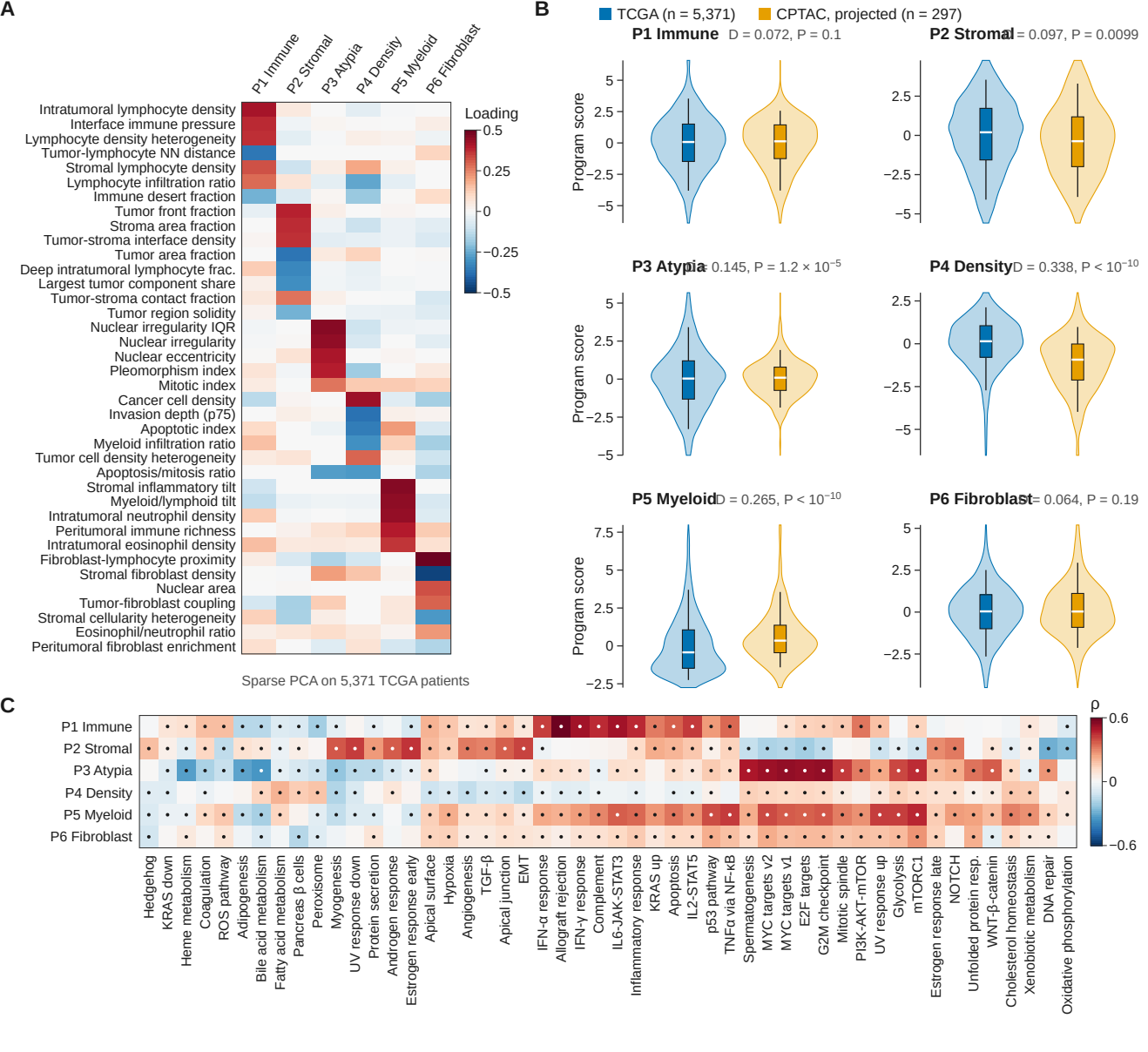

A
P1 Immune
P2 Stromal
P3 Atypia
P4 Density
P5 Myeloid
P6 Fibroblast
Loading
0.5
0.25
0
−0.25
−0.5
Intratumoral lymphocyte density
Interface immune pressure
Lymphocyte density heterogeneity
Tumor-lymphocyte NN distance
Stromal lymphocyte density
Lymphocyte infiltration ratio
Immune desert fraction
Tumor front fraction
Stroma area fraction
Tumor-stroma interface density
Tumor area fraction
Deep intratumoral lymphocyte frac.
Largest tumor component share
Tumor-stroma contact fraction
Tumor region solidity
Nuclear irregularity IQR
Nuclear irregularity
Nuclear eccentricity
Pleomorphism index
Mitotic index
Cancer cell density
Invasion depth (p75)
Apoptotic index
Myeloid infiltration ratio
Tumor cell density heterogeneity
Apoptosis/mitosis ratio
Stromal inflammatory tilt
Myeloid/lymphoid tilt
Intratumoral neutrophil density
Peritumoral immune richness
Intratumoral eosinophil density
Fibroblast-lymphocyte proximity
Stromal fibroblast density
Nuclear area
Tumor-fibroblast coupling
Stromal cellularity heterogeneity
Eosinophil/neutrophil ratio
Peritumoral fibroblast enrichment
Sparse PCA on 5,371 TCGA patients
B
TCGA (n = 5,371)
CPTAC, projected (n = 297)
P1 Immune D = 0.072, P = 0.1
P2 Stromal D = 0.097, P = 0.0099
P3 Atypia D = 0.145, P = 1.2 × 10−5
P4 Density D = 0.338, P < 10−10
P5 Myeloid D = 0.265, P < 10−10
P6 Fibroblast D = 0.064, P = 0.19
Program score
C
ρ
0.6
0
−0.6
P1 Immune
P2 Stromal
P3 Atypia
P4 Density
P5 Myeloid
P6 Fibroblast
Hedgehog
KRAS down
Heme metabolism
Coagulation
ROS pathway
Adipogenesis
Bile acid metabolism
Fatty acid metabolism
Pancreas β cells
Peroxisome
Myogenesis
UV response down
Protein secretion
Androgen response
Estrogen response early
Apical surface
Hypoxia
Angiogenesis
TGF-β
Apical junction
EMT
IFN-α response
Allograft rejection
IFN-γ response
Complement
IL6-JAK-STAT3
Inflammatory response
KRAS up
Apoptosis
IL2-STAT5
p53 pathway
TNFα via NF-κB
Spermatogenesis
MYC targets v2
MYC targets v1
E2F targets
G2M checkpoint
Mitotic spindle
PI3K-AKT-mTOR
UV response up
Glycolysis
mTORC1
Estrogen response late
NOTCH
Unfolded protein resp.
WNT-β-catenin
Cholesterol homeostasis
Xenobiotic metabolism
DNA repair
Oxidative phosphorylation